Financial Econometrics

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Efficient portfolios for the mean-variance criterion

Portfolio theory

- Aims at determining which combinations of assets have an optimal return at a given risk level (or a minimal risk at a given return level).
- The mean-variance variance analysis, in which return and risk are measured via the **first and second moments** of the returns, is central in the portfolio theory.
- More than 60 years after its introduction [Markowitz (1952)] it remains - despite its simplicity - the more widely used allocation method.

Portfolio theory

Several reasons for this interest :

- Provides an intuitive interpretation of the performance of portfolios, which can be visualized on a graph.
- Efficient frontiers are generated by only two funds. This property simplifies the analysis and leads to the CAPM (Capital Asset Pricing Model) by Sharpe (1964), Lintner (1965) and Mossin (1966).
- The mean-variance approach is natural for Gaussian and elliptic distributions. It is compatible with the maximisation of utility whatever the preferences of the agents (see Chamberlain (1983), Owen and Rabonovitch (1983) and Berk (1997)).

Assumptions

Consider an economy with 1 risk-free asset and N risky assets. By convention, the risk-free asset is priced 1 at time t. Let

- r_t the risk-free rate between t and $t+1$,
- $p_{i,t}$ **the price of asset i at time t** ($i = 1, \ldots, N$) and let $r_{i,t+1}$ its net return between t and $t+1$,

$$r_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}} - 1 := y_{i,t+1} - 1.$$

Let $\mathbf{y}_{t+1} = (y_{i,t+1})_{i=1,\ldots,N}$.
Assume that

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}_t, \qquad \mathrm{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}_t,$$

where $\boldsymbol{\mu}_t = (\mu_{1t}, \ldots, \mu_{Nt})' \in \mathbb{R}^N$ and $\boldsymbol{\Omega}_t$ is a $N \times N$ positive definite matrix. We assume that $\mu_{it} \geq 1 + r_t$, for $i = 1, \ldots, N$.

Conditional moments of the portfolio's value

Let a portfolio : a_0 units of the risk-free asset and a_i units of asset i, for $i=1,\ldots,N$.
Value at time t :

$$V_t = a_0 + \mathbf{a}'\mathbf{p}_t, \quad \mathbf{a} = (a_1,\ldots,a_N)', \quad \mathbf{p}_t = (p_{1,t},\ldots,p_{N,t})'.$$

Suppose the portfolio's composition is fixed between t and $t+1$.

$$\begin{aligned} V_{t+1} &= a_0(1+r_t) + \sum_{i=1}^{N} a_i p_{i,t} y_{i,t+1} \\ &= a_0(1+r_t) + \mathbf{a}'\text{diag}(\mathbf{p}_t)\mathbf{y}_{t+1} \end{aligned}$$

where diag($\mathbf{p}_t$) is the diagonal matrix whose diagonal elements are the $p_{i,t}$. The expected return and variability of this portfolio are

$$\begin{aligned} E_t V_{t+1} &= a_0(1+r_t) + \mathbf{a}'\text{diag}(\mathbf{p}_t)\boldsymbol{\mu}_t, \\ \text{Var}_t(V_{t+1}) &= \mathbf{a}'\text{diag}(\mathbf{p}_t)\boldsymbol{\Omega}_t\text{diag}(\mathbf{p}_t)\mathbf{a}. \end{aligned}$$

A preorder on the portfolios

The *mean-variance criterion* consists in selecting portfolios which maximize the expected return and minimize the variability (risk).

More precisely, we define a preorder * on portfolios.

Definition

Let two portfolios P and $\tilde{P}$, which have the same value at time t and respective values V_{t+1} and $\tilde{V}_{t+1}$ at $t+1$. Portfolio P is called **preferable** *to $\tilde{P}$ if*

$$E_t V_{t+1} \geq E_t \tilde{V}_{t+1} \quad \text{and} \quad Var_t V_{t+1} \leq Var_t \tilde{V}_{t+1}.$$

The efficient frontier is defined as the set of portfolios which are not dominated in this preorder.

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*. Reflexive and transitive (though not antisymmetric).

Optimization problem

Let $V_t = V_t(a_0, \mathbf{a})$.

For a given variance level σ_t^2, the mean-variance optimization program is

$$\begin{cases} \max_{a_0,\mathbf{a}} \; E_t V_{t+1}(a_0,\mathbf{a}) \\ \\ \text{subject to} \quad \mathrm{Var}_t V_{t+1}(a_0,\mathbf{a}) = \sigma_t^2, \end{cases}$$

for portfolios of initial value $V_t(a_0,\mathbf{a}) = v$.

The program writes

$$\begin{cases} \max_{a_0,\mathbf{a}} \; a_0(1+r_t) + \mathbf{a}'\mathrm{diag}(\mathbf{p}_t)\boldsymbol{\mu}_t \\ \\ \text{subject to} \quad \mathbf{a}'\mathrm{diag}(\mathbf{p}_t)\boldsymbol{\Omega}_t\mathrm{diag}(\mathbf{p}_t)\mathbf{a} = \sigma_t^2. \end{cases}$$

Equivalent formulation

Let

$$\boldsymbol{\omega}_{\mathbf{a}} = \mathrm{diag}(\mathbf{p}_t)\mathbf{a}.$$

Since

$$v = a_0 + \boldsymbol{\omega}_{\mathbf{a}}'\mathbf{e}, \quad \mathbf{e} = (1, \ldots, 1)',$$

we have

$$\begin{cases} \max_{\boldsymbol{\omega}_{\mathbf{a}}} \ \{v - \boldsymbol{\omega}_{\mathbf{a}}'\mathbf{e}\}(1 + r_t) + \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\mu}_t \\ \\ \text{subject to} \quad \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}_t\boldsymbol{\omega}_{\mathbf{a}} = \sigma_t^2 \end{cases}$$

or equivalently

$$\begin{cases} \max_{\boldsymbol{\omega}_{\mathbf{a}}} \ \boldsymbol{\omega}_{\mathbf{a}}'\{\boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}\} \\ \\ \text{subject to} \quad \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}_t\boldsymbol{\omega}_{\mathbf{a}} = \sigma_t^2 \end{cases}$$

First-order condition

Let

$$\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}.$$

Introducing the Lagrangian function, the programme writes

$$\begin{cases} \max_{\boldsymbol{\omega}_{\mathbf{a}},\lambda} \mathscr{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda) \\ \\ \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda) = \boldsymbol{\omega}_{\mathbf{a}}' \tilde{\boldsymbol{\mu}}_t - \frac{\lambda}{2}\{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} - \sigma_t^2\}. \end{cases}$$

The first-order conditions are

$$\begin{cases} \frac{\partial \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda)}{\partial \boldsymbol{\omega}_{\mathbf{a}}} = \tilde{\boldsymbol{\mu}}_t - \lambda \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} = 0 \\ \\ \frac{\partial \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda)}{\partial \lambda} = -\frac{1}{2}\{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} - \sigma_t^2\} = 0. \end{cases}$$

Solution

$$\Rightarrow \begin{cases} \boldsymbol{\omega}_{\mathbf{a}} = \frac{1}{\lambda}\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t, \\ \\ \lambda^2 = \frac{1}{\sigma_t^2}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t. \end{cases}$$

For such portfolios we have :

$$\begin{aligned} E_t(V_{t+1}) &= v(1+r_t) + \boldsymbol{\omega}_{\mathbf{a}}'\tilde{\boldsymbol{\mu}}_t \\ &= v(1+r_t) + \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t. \end{aligned}$$

Thus it is the solution $\lambda \geq 0$ which corresponds to a maximum †.

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†. Otherwise, investing all initial capital in the riskless asset would have a better return, $v(1+r_t)$, without any risk.

Characterization of the efficient frontier

Proposition

For portfolios with value v at time t, the efficient frontier is the set of portfolios such that

$$\begin{cases} \mathbf{a} = \frac{1}{\lambda}\{diag(\mathbf{p}_t)\}^{-1}\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t, & \lambda \geq 0, \\ a_0 = v - \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}e, & \\ \lambda = \frac{1}{\sigma_t}\left\{\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t\right\}^{1/2}. & \end{cases}$$

Expected return of efficient portfolios :

$$E_t(V_{t+1}) \quad = \quad v(1+r_t) + \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t.$$

Remarks :

- The initial value v of the portfolio only appears in a_0.
  Efficient frontiers corresponding to different values of v are obtained by vertical translation.
- Some coefficients a_i can be negative.
  Can be interpreted as a "short" position on the corresponding assets (the investor borrows those assets to a broker and sells them, before buying them at an ulterior date to give them back to the broker).
- If $\boldsymbol{\Omega}_t$ is diagonal, the a_i are positive because the components of $\tilde{\boldsymbol{\mu}}_t$ are positive.
- If short positions are not allowed, constraints $a_i \geq 0$ should be included in the program. No explicit solution in general.
- The multiplicator λ varies as the inverse of σ_t. Can be interpreted as a measure of the *risk aversion* of the investor.

Remarks :

- The efficient frontier can be deduced from other optimization problems. For instance, consider the dual program :

$$\begin{cases} \min_{a_0,\mathbf{a}} \mathrm{Var}_t V_{t+1}(a_0,\mathbf{a}) \\ \text{subject to} \quad E_t V_{t+1}(a_0,\mathbf{a}) = m_t, \end{cases}$$

  for portfolios with initial value fixed at $V_t(a_0,\mathbf{a}) = v$.
  Same solution if m_t and σ_t are related by :

$$m_t = v(1+r_t) + \sigma_t \left\{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t\right\}^{1/2}$$

- An alternative approach is based on the maximization of the expected utility of the future wealth. The equivalence between the two approaches is obtained under severe restrictions on the utility function (Samuelson (1970), Levy and Markowitz (1979), Epstein (1985)) or on the law of the returns (Chamberlain, 1983).

Separation in two funds

Every efficient portfolio can be written as

$$\begin{aligned}\begin{pmatrix} a_0 \\ \mathbf{a} \end{pmatrix} &= \begin{pmatrix} v \\ \mathbf{0} \end{pmatrix} + \frac{1}{\lambda}\begin{pmatrix} -\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\mathbf{e} \\ \{\mathrm{diag}(\mathbf{p}_t)\}^{-1}\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t \end{pmatrix} \\ &:= \mathbf{P_0} + \frac{1}{\lambda}\mathbf{P}^*,\end{aligned}$$

$\mathbf{P_0}$ is solely invested in the risk-free asset.

$\mathbf{P}^*$ is called tangent portfolio : short positions finance long positions (null value at t).

This decomposition is known as *two funds separation theorem*.

Optimal portfolio

For an investor, an optimal portfolio can be obtained in 2 steps :

- computing the tangent portfolio (that is, the optimal portfolio invested in the risky assets) ;
- choosing a repartition between the riskless portfolio and the tangent portfolio depending on the agent's risk aversion.

Portfolios can be visualized on a graph : each portfolio is represented by its risk $\mathrm{Var}_t(V_{t+1})$ (horizontal axis) and its anticipated value $E_t(V_{t+1})$ (vertical axis).

Portfolios which are preferable to a given portfolio appear in its North-West quadrant.

Parameterization for efficient portfolios

For an efficient portfolio, the expected value and risk are thus parameterized by the coefficient λ :

$$\begin{aligned} m_t(\lambda) := E_t V_{t+1} &= v(1+r_t) + \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t, \\ \sigma_t^2(\lambda) := \mathrm{Var}_t V_{t+1} &= \frac{1}{\lambda^2}\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t. \end{aligned}$$

By eliminating λ we get

$$\sigma_t^2(\lambda) = \frac{\{m_t(\lambda) - (1+r_t)v\}^2}{\tilde{\boldsymbol{\mu}}_t'\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t}.$$

Efficient frontier

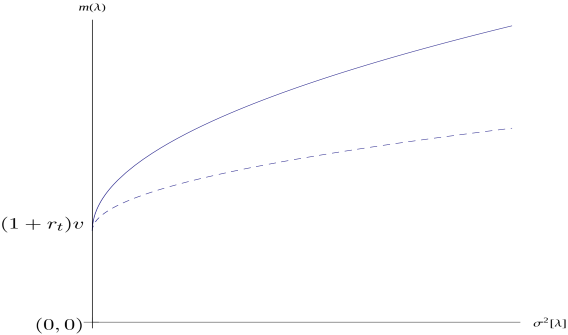
In the mean-variance diagram, the efficient frontier is a semiparabola with vertex $(\sigma_t^2(\lambda) = 0, m_t(\lambda) = (1+r_t)v)$, corresponding to a portfolio which is fully invested in the risk-free asset.

In this diagram, points below the efficient frontier correspond to non-efficient portfolios.

Each point of the semiparabola corresponds to a degree of risk aversion.
($\lambda = \infty$ for the riskless portfolio, $\lambda \to 0$ for the asymptote).

Efficient frontier (full line)

Example of portfolios with the same performance (dotted line)



Optimization program in the absence of riskfree asset

Without a riskfree asset, the optimization program becomes

$$\begin{cases} \max_{\boldsymbol{\omega}_{\mathbf{a}}} \ \boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\mu}_t \\ \\ \text{subject to} \quad \boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} = \sigma_t^2, \quad v = \boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{e}. \end{cases}$$

Introducing the Lagrangian, the program writes

$$\begin{cases} \max_{\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2} \mathscr{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2) \\ \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2) = \boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\mu}_t - \frac{\lambda_1}{2}(\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} - \sigma_t^2) - \lambda_2(\boldsymbol{\omega}_{\mathbf{a}}' \mathbf{e} - v). \end{cases}$$

The first-order conditions write

$$\begin{cases} \frac{\partial \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2)}{\partial \boldsymbol{\omega}_{\mathbf{a}}} = \boldsymbol{\mu}_t - \lambda_1 \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} - \lambda_2 \mathbf{e} = 0, \\ \frac{\partial \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2)}{\partial \lambda_1} = -\frac{1}{2}(\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}} - \sigma_t^2) = 0, \\ \frac{\partial \mathbf{L}(\boldsymbol{\omega}_{\mathbf{a}},\lambda_1,\lambda_2)}{\partial \lambda_2} = v - \boldsymbol{\omega}_{\mathbf{a}}' \mathbf{e} = 0. \end{cases}$$

Optimization program in the absence of riskfree asset

Thus

$$\boldsymbol{\omega}_{\mathbf{a}} = \frac{1}{\lambda_1}\boldsymbol{\Omega}_t^{-1}(\boldsymbol{\mu}_t - \lambda_2 \mathbf{e}).$$

Plugging this equality in the previous relations, we get

$$\lambda_2 = \frac{\boldsymbol{\mu}_t' \boldsymbol{\Omega}_t^{-1}\mathbf{e} - \lambda_1 v}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}},$$

$$\lambda_1^2 = \frac{1}{\sigma_t^2}(\boldsymbol{\mu}_t - \lambda_2\mathbf{e})'\boldsymbol{\Omega}_t^{-1}(\boldsymbol{\mu}_t - \lambda_2\mathbf{e}).$$

Optimal allocation in the absence of riskfree asset

Therefore, the optimal allocation is

$$\boldsymbol{\omega}_{\mathbf{a}} = \frac{1}{\lambda_1}\boldsymbol{\Omega}_t^{-1}\left(\boldsymbol{\mu}_t - \frac{\boldsymbol{\mu}_t'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\mathbf{e}\right) + \frac{v}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\boldsymbol{\Omega}_t^{-1}\mathbf{e},$$

with

$$\lambda_1 = \frac{1}{\sigma_t}\sqrt{(\boldsymbol{\mu}_t - \lambda_2\mathbf{e})'\boldsymbol{\Omega}_t^{-1}(\boldsymbol{\mu}_t - \lambda_2\mathbf{e})}.$$

Note that the initial wealth is involved in the allocation.

Remark :

$$\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\mu}_t = v\frac{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\boldsymbol{\mu}_t}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}} + \frac{1}{\lambda_1}\left(\boldsymbol{\mu}_t'\boldsymbol{\Omega}_t^{-1}\boldsymbol{\mu}_t - \frac{(\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\boldsymbol{\mu}_t)^2}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\right).$$

Thus, if $v=0$, the expected value of the portfolio at $t+1$ is positive :

$$\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\mu}_t = \frac{1}{\lambda_1}\left(\boldsymbol{\mu}_t'\boldsymbol{\Omega}_t^{-1}\boldsymbol{\mu}_t - \frac{(\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\boldsymbol{\mu}_t)^2}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\right) \geq 0.$$

Efficient frontier in the absence of riskfree asset

Let $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - \frac{\boldsymbol{\mu}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \mathbf{e}} \mathbf{e}$.

We have

$$\lambda_1 = \frac{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}{m_t - v \frac{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \boldsymbol{\mu}_t}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \mathbf{e}}}.$$

Thus

$$\sigma_t^2 = \frac{v^2}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \mathbf{e}} + \frac{\left(m_t - v \frac{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \boldsymbol{\mu}_t}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \mathbf{e}} \right)^2}{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}.$$

The efficient frontier is a semi-parabola in the plan (σ_t^2, m_t).

Substitute of the risk-free asset

The minimal-risk efficient portfolio is such that $\lambda_1 = +\infty$.
Thus its composition is

$$\boldsymbol{\omega}_{\mathbf{a}^*} = \frac{v}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\boldsymbol{\Omega}_t^{-1}\mathbf{e}.$$

The efficient frontier is generated by the portfolios $\boldsymbol{\omega}_{\mathbf{a}^*}$ and

$$\boldsymbol{\omega}_{\mathbf{a}^{**}} = \boldsymbol{\Omega}_t^{-1}\left(\boldsymbol{\mu}_t - \frac{\boldsymbol{\mu}_t'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}{\mathbf{e}'\boldsymbol{\Omega}_t^{-1}\mathbf{e}}\mathbf{e}\right)$$

since

$$\boldsymbol{\omega}_{\mathbf{a}} = \boldsymbol{\omega}_{\mathbf{a}^*} + \frac{1}{\lambda_1}\boldsymbol{\omega}_{\mathbf{a}^{**}}.$$

Portfolio $\boldsymbol{\omega}_{\mathbf{a}^*}$ plays the role of the risk-free asset in the standard model.
Once this effect is eliminated (or equivalently, when $v = 0$) all efficient portfolios are proportional to $\boldsymbol{\omega}_{\mathbf{a}^{**}}$.

Sharpe Performance

The curvature of the efficient frontier is related to the *Sharpe performance.*

Definition

Sharpe performance (or market performance) of the set of all assets is defined as

$$S_t = \sqrt{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}.$$

When $\boldsymbol{\Omega}_t$ is a diagonal matrix, we have

$$S_t = \sqrt{\sum_{i=1}^{N} \frac{(\mu_{it} - (1 + r_t))^2}{\sigma_{it}^2}}$$

where μ_{it} and σ_{it}^2 are the conditional mean and variance of asset i.

Sharpe ratio

Sharpe ratio of a portfolio :

$$S_t(\mathbf{a}) = \frac{E_t V_{t+1}(a_0, \mathbf{a}) - (1 + r_t)v}{\sqrt{\mathrm{Var}_t\{V_{t+1}(a_0, \mathbf{a})\}}}.$$

We have

$$S_t(\mathbf{a}) = \frac{\boldsymbol{\omega}_{\mathbf{a}}' \tilde{\boldsymbol{\mu}}_t}{(\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}})^{1/2}}.$$

Portfolios with the same Sharpe ratio : a semi-parabola, intersecting the efficient frontier at the vertical axis but with a different curvature.

Proposition

For any portfolio, $S_t(\mathbf{a}) \leq S_t$. A portfolio is efficient iff $S_t(\mathbf{a}) = S_t$.

This property is used for testing the efficiency of a portfolio

Proof

Since $\boldsymbol{\Omega}_t$ is positive-definite, there exists a positive-definite matrix $\mathbf{R}_t$ such that $\boldsymbol{\Omega}_t = \mathbf{R}_t^2$. [‡] Let $\mathbf{R}_t = \boldsymbol{\Omega}_t^{1/2}$. We have

$$S_t(\mathbf{a}) = \frac{\boldsymbol{\omega}_{\mathbf{a}}' \tilde{\boldsymbol{\mu}}_t}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}}}} = \frac{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t^{1/2} \boldsymbol{\Omega}_t^{-1/2} \tilde{\boldsymbol{\mu}}_t}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}}}} \leq \sqrt{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}$$

by Cauchy-Schwarz inequality, which proves the inequality.

It also follows that the market performance is the Sharpe ratio of the efficient portfolios.

$\square$

‡. For instance $\mathbf{R}_t = P_t \Lambda_t^{1/2} P_t'$, where $\Lambda_t^{1/2}$ is diagonal with the square-roots of the eigenvalues of $\boldsymbol{\Omega}_t$ on the diagonal, and P_t is the orthogonal matrix of eigenvectors.

Another expression of the Sharpe performance

$S_t(e_i)$: Sharpe ratio of the portfolio built with the ith asset.

$\mathbf{S}_t = (S_t(e_i)) \in \mathbb{R}^N$.

Let $\boldsymbol{\rho}_t$ the conditional correlation matrix :

$$\boldsymbol{\rho}_t = \mathrm{diag}(\boldsymbol{\Omega}_t)^{-1/2}\boldsymbol{\Omega}_t \mathrm{diag}(\boldsymbol{\Omega}_t)^{-1/2}$$

Proposition

The market performance is related to the Sharpe performance of the N risky assets by

$$S_t^2 = \mathbf{S}'_t \boldsymbol{\rho}_t^{-1} \mathbf{S}_t.$$

In particular, if the assets are uncorrelated, the market performance reduces to the square-root of the sum of the squared performances of the N assets.

Proof

It suffices to note that

$$S_t(e_i) = \frac{\mu_{it} - (1 + r_t)}{\sigma_{it}}.$$

Thus

$$\mathbf{S}_t = \mathrm{diag}(\boldsymbol{\Omega}_t)^{-1/2}\tilde{\boldsymbol{\mu}}_t$$

and the conclusion follows.

$\square$

Example

For $N = 2$ risky assets, we get a more explicit formula.

Let $\rho_t = \text{Cor}_t(y_{1,t+1}, y_{2,t+1})$.

The market performance is given by

$$S_t^2 = \frac{1}{1-\rho_t^2}\{S_t^2(e_1) - 2\rho_t S_t(e_1) S_t(e_2) + S_t^2(e_2)\}.$$

Link with the regression

In the multiple regression of a variable Y on a constant and a vector $\boldsymbol{X}$, the theoretical determination coefficient is given by

$$R^2 = \boldsymbol{\rho}_{Y\boldsymbol{X}} \boldsymbol{\rho}_{\boldsymbol{XX}}^{-1} \boldsymbol{\rho}_{\boldsymbol{X}Y}$$

where $\boldsymbol{\rho}_{\boldsymbol{XX}}$ is the correlation matrix of $\boldsymbol{X}$ and $\boldsymbol{\rho}_{\boldsymbol{X}Y}$ is the vector of correlations between Y and the components of $\boldsymbol{X}$.

In the regression of any portfolio V_{t+1} on $\mathbf{y}_{t+1}$, the R^2 is equal to 1 because V_{t+1} and $\mathbf{y}_{t+1}$ are colinear.

Link with the regression

Efficient portfolio :

$$V^*_{t+1} = a_0(1+r_t) + \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}'_t \boldsymbol{\Omega}_t^{-1} \boldsymbol{y}_{t+1}.$$

$$\mathsf{Cov}_t(V^*_{t+1}, \boldsymbol{y}_{t+1}) = \frac{1}{\lambda}\tilde{\boldsymbol{\mu}}'_t, \quad \mathsf{Var}_t(V^*_{t+1}) = \frac{1}{\lambda^2}\tilde{\boldsymbol{\mu}}'_t \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t,$$

$$\Rightarrow \mathsf{Corr}_t\left(V^*_{t+1}, y_{i,t+1}\right) = \frac{\frac{1}{\lambda}\tilde{\boldsymbol{\mu}}_{it}}{\frac{1}{\lambda}\sqrt{\tilde{\boldsymbol{\mu}}'_t \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}\,\sigma_{it}} = \frac{\tilde{\boldsymbol{\mu}}_{it}}{\sigma_{it}}\frac{1}{S_t} = \frac{S_t(e_i)}{S_t},$$

$$\Rightarrow \mathsf{Corr}_t\left(V^*_{t+1}, \boldsymbol{y}_{t+1}\right) = \frac{\mathbf{S}'_t}{S_t},$$

$$\Rightarrow R^2 = 1 = \frac{\mathbf{S}'_t}{S_t}\boldsymbol{\rho}_t^{-1}\frac{\mathbf{S}_t}{S_t}.$$

$\square$

Estimation of $\boldsymbol{\mu}_t$ and $\boldsymbol{\Omega}_t$

Assume that

$$E_t(\mathbf{y}_{t+1}) = \boldsymbol{\mu}, \quad \mathsf{Var}_t(\mathbf{y}_{t+1}) = \boldsymbol{\Omega}, \qquad \forall t.$$

If the returns are stationary, these theoretical moments can be estimated from a time series $(\mathbf{y}_t)_{t=1,\dots,n}$, by taking the empirical moments :

$$\hat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{t=1}^{n}\mathbf{y}_t, \qquad \hat{\boldsymbol{\Omega}}_n = \frac{1}{n}\sum_{t=1}^{n}(\mathbf{y}_t - \hat{\boldsymbol{\mu}}_n)(\mathbf{y}_t - \hat{\boldsymbol{\mu}}_n)'.$$

Asymptotic properties of the empirical moments

Proposition

Assume that the variables $\mathbf{y}_t$, $t \in \mathbb{Z}$, are i.i.d. Then

1. $\hat{\boldsymbol{\mu}}_n \to \boldsymbol{\mu}, \quad \hat{\boldsymbol{\Omega}}_n \to \boldsymbol{\Omega}, \quad a.s. \quad$ *and* $\quad \sqrt{n}(\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Omega})$.
2. *If, in addition,* $\mathbf{y}_t \rightsquigarrow \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$,
   1. $\hat{\boldsymbol{\mu}}_n$ *and* $\hat{\boldsymbol{\Omega}}_n$ *are asymptotically Gaussian and independent,*
   2. *the asymptotic law of* $vech\{\sqrt{n}(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\}$ *is Gaussian, centered, with asymptotic covariance matrix characterized by :* $\forall x, y \in \mathbb{R}^N$,

$$Cov_{as}\{\sqrt{n}(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})x, \sqrt{n}(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})y\} = (\mathbf{x}'\boldsymbol{\Omega}y).\boldsymbol{\Omega} + \boldsymbol{\Omega}y\mathbf{x}'\boldsymbol{\Omega}.$$

Remarks

1. The first two convergences follow from the SLLN. The convergence in distribution of $\hat{\boldsymbol{\mu}}_n$ follows from the CLT. The convergence in law of $\hat{\boldsymbol{\Omega}}_n$ in the Gaussian case is established in Anderson (1984).
2. The last result is often used to show that for any non-zero vector $x \in \mathbb{R}^N$,

$$
\begin{aligned}
\sqrt{n}x'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})x &\overset{d}{\to} \mathcal{N}\left(0, 2(x'\boldsymbol{\Omega}x)^2\right), \\
\sqrt{n}x'(\hat{\boldsymbol{\Omega}}_n^{-1} - \boldsymbol{\Omega}^{-1})x &\overset{d}{\to} \mathcal{N}\left(0, 2(x'\boldsymbol{\Omega}^{-1}x)^2\right).
\end{aligned}
$$

3. The Gaussian assumption is an approximation because the components of $\mathbf{y}_t$ are positive (price ratios).

Estimation of the Sharpe ratio

The Sharpe ratio at time t of a portfolio $\mathbf{a}$ can be estimated by

$$\hat{S}_{t,n}(\mathbf{a}) = \frac{\boldsymbol{\omega}_{\mathbf{a}}'\tilde{\boldsymbol{\mu}}_n}{\{\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}}\}^{1/2}}, \qquad \tilde{\boldsymbol{\mu}}_n = \hat{\boldsymbol{\mu}}_n - (1+r_t)\mathbf{e}.$$

We assume that observations anterior to time 0, $(\mathbf{y}_{-t})_{1\leq t\leq n}$ are used to compute

$$\hat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{t=1}^{n}\mathbf{y}_{-t}, \quad \hat{\boldsymbol{\Omega}}_n = \frac{1}{n}\sum_{t=1}^{n}(\mathbf{y}_{-t}-\hat{\boldsymbol{\mu}}_n)(\mathbf{y}_{-t}-\hat{\boldsymbol{\mu}}_n)'.$$

Let $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - (1+r_t)\mathbf{e}$.

Assumptions

A0 : $(\mathbf{y}_t) \overset{iid}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ where $\boldsymbol{\mu} \in \mathbb{R}^N$ and $\boldsymbol{\Omega}$ is a $(N \times N)$ definite positive matrix. Moreover,

$$\boldsymbol{\mu}_t = \boldsymbol{\mu} \quad \text{and} \quad \boldsymbol{\Omega}_t = \boldsymbol{\Omega}.$$

A1 : The statistics $\hat{\boldsymbol{\mu}}_n$ and $\hat{\boldsymbol{\Omega}}_n$ are asymptotically independent of $\mathbf{p}_t$ for $t > 0$.

Comments

$$\boldsymbol{\mu}_t = E_t(\mathbf{y}_{t+1}) = \begin{pmatrix} E_t(p_{1,t+1})/p_{1t} \\ \vdots \\ E_t(p_{N,t+1})/p_{Nt} \end{pmatrix}, \quad \boldsymbol{\mu} = E(\mathbf{y}_{t+1}) = \begin{pmatrix} E(p_{1,t+1}/p_{1t}) \\ \vdots \\ E(p_{N,t+1}/p_{Nt}) \end{pmatrix}.$$

A model which is compatible with **A0** is the random walk on the log-prices :

$$\log p_{i,t+1} = \log p_{i,t} + \epsilon_{i,t+1}, \quad i = 1, \ldots, N$$

where $(\boldsymbol{\epsilon}_t)$ is an iid sequence, with $\boldsymbol{\epsilon}_{t+1} = (\epsilon_{i,t+1})_{i=1,\ldots,N}$ independent of $(\log p_{i,t})_{i=1,\ldots,N}$.

Asymptotic behavior of $\hat{S}_{t,n}(\mathbf{a})$

Proposition

Under **A0**-**A1**, *we have, conditional on* $\mathbf{p}_t$,

$$\hat{S}_{t,n}(\mathbf{a}) \to S_t(\mathbf{a}), \quad a.s.$$

$$\sqrt{n}\{\hat{S}_{t,n}(\mathbf{a}) - S_t(\mathbf{a})\} \xrightarrow{d} \mathcal{N}\left(0, 1 + \frac{1}{2} S_t^2(\mathbf{a})\right).$$

Note that the asymptotic accuracy of the estimator only depends on the performance of the portfolio.

The price $\mathbf{p}_t$ is involved in the asymptotic variance through $\boldsymbol{\omega}_{\mathbf{a}}$.

Proof (I)

The consistency follows from the consistency of the empirical moments.

We have

$$\sqrt{n}\boldsymbol{\omega}_{\mathbf{a}}'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{a}} \xrightarrow{d} \mathcal{N}(0, 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2).$$

$\hat{S}_{t,n}(\mathbf{a})$ is a differentiable function f of $\hat{\boldsymbol{\mu}}_n$ and $\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}}$, which are asymptotically independent.

Proof (II)

By the delta method,

$$
\begin{aligned}
& \sqrt{n}(\hat{S}_{t,n}(\mathbf{a}) - S_t(\mathbf{a})) \\
= \quad & \sqrt{n}\{f(\tilde{\boldsymbol{\mu}}_n, \boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}}) - f(\tilde{\boldsymbol{\mu}}, \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})\}, \qquad f(\boldsymbol{\mu}, z) = \frac{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\mu}}{\sqrt{z}}, \\
\xrightarrow{d} \quad & \mathbf{N}\left(0, \left[\frac{\partial f}{\partial \boldsymbol{\mu}'} \quad \frac{\partial f}{\partial z}\right]_{z=\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}} \begin{pmatrix} \boldsymbol{\Omega} & 0 \\ 0 & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2 \end{pmatrix} \left[\frac{\partial f}{\partial \boldsymbol{\mu}'} \quad \frac{\partial f}{\partial z}\right]'_{z=\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}\right),
\end{aligned}
$$

and we conclude using

$$
\begin{aligned}
\left[\frac{\partial f}{\partial \boldsymbol{\mu}'} \quad \frac{\partial f}{\partial z}\right]_{z=\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}} &= \left[\frac{\boldsymbol{\omega}_{\mathbf{a}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}}, \quad \frac{-\boldsymbol{\omega}_{\mathbf{a}}'\tilde{\boldsymbol{\mu}}}{2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^{3/2}}\right] \\
&= \left[\frac{\boldsymbol{\omega}_{\mathbf{a}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}}, \quad \frac{-S_t(\mathbf{a})}{2\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}\right].
\end{aligned}
$$

$\square$

Consequence

The market performance is the Sharpe ratio of an efficient portfolio with composition $\boldsymbol{\omega}_{\mathbf{a}^*} = \boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}$ (obtained for $\lambda = 1$).

It follows that

$$\sqrt{n}\{\hat{S}_{t,n}(\mathbf{a}^*) - S_t)\} \xrightarrow{d} \mathcal{N}\left(0, 1 + \frac{1}{2}S_t^2\right).$$

This result, however, has little practical interest because $\mathbf{a}^*$ depends on unknown parameters : hence $\hat{S}_{t,n}(\mathbf{a}^*)$ is not an estimator of S_t.

Asymptotic behavior of $\hat{S}_{t,n}$

A natural estimator of the market performance is

$$\hat{S}_{t,n} = \sqrt{\tilde{\boldsymbol{\mu}}_n' \hat{\boldsymbol{\Omega}}_n^{-1} \tilde{\boldsymbol{\mu}}_n}.$$

Proposition

Under **A0**-**A1**, *we have, conditional on* $\mathbf{p}_t$,

$$\hat{S}_{t,n} \to S_t, \quad a.s., \qquad \sqrt{n}\{\hat{S}_{t,n} - S_t\} \xrightarrow{d} \mathcal{N}\left(0, 1 + \frac{1}{2} S_t^2\right).$$

Note that the asymptotic law is the same as that of $\hat{S}_{t,n}(\mathbf{a}^*)$.

Proof (I)

We have

$$\sqrt{n}\{\hat{S}_{t,n}-S_t\}=\sqrt{n}\frac{\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n-\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}}{\sqrt{\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}}+\sqrt{\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}}},$$

with

$$\begin{aligned}
& \sqrt{n}(\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n-\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}) \\
= \; & \sqrt{n}\tilde{\boldsymbol{\mu}}'(\hat{\boldsymbol{\Omega}}_n^{-1}-\boldsymbol{\Omega}^{-1})\tilde{\boldsymbol{\mu}}+\sqrt{n}(\tilde{\boldsymbol{\mu}}_n'-\tilde{\boldsymbol{\mu}}')\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n+\tilde{\boldsymbol{\mu}}'\hat{\boldsymbol{\Omega}}_n^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n-\tilde{\boldsymbol{\mu}}) \\
\overset{d}{=} \; & \sqrt{n}\tilde{\boldsymbol{\mu}}'(\hat{\boldsymbol{\Omega}}_n^{-1}-\boldsymbol{\Omega}^{-1})\tilde{\boldsymbol{\mu}}+2\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n-\tilde{\boldsymbol{\mu}})
\end{aligned}$$

These two quantities are asymptotically independent, with

$$2\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n-\tilde{\boldsymbol{\mu}})\xrightarrow{d}\mathcal{N}\left(0,4\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}\right),$$

$$\sqrt{n}\tilde{\boldsymbol{\mu}}'(\hat{\boldsymbol{\Omega}}_n^{-1}-\boldsymbol{\Omega}^{-1})\tilde{\boldsymbol{\mu}} \quad \xrightarrow{d} \quad \mathcal{N}\left(0,2(\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}})^2\right).$$

Proof (II)

Thus

$$\sqrt{n}(\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n - \tilde{\boldsymbol{\mu}}'\hat{\boldsymbol{\Omega}}^{-1}\tilde{\boldsymbol{\mu}}) \xrightarrow{d} \mathcal{N}\left(0, 4\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}} + 2(\tilde{\boldsymbol{\mu}}'\boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}})^2\right)$$

Because

$$\sqrt{\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}} + \sqrt{\tilde{\boldsymbol{\mu}}'\hat{\boldsymbol{\Omega}}^{-1}\tilde{\boldsymbol{\mu}}} \xrightarrow{P} 2\sqrt{\tilde{\boldsymbol{\mu}}'\hat{\boldsymbol{\Omega}}^{-1}\tilde{\boldsymbol{\mu}}},$$

we get the result.

$\square$

Asymptotic behavior of the estimator of an optimal portfolio

The optimal allocation is proportional to

$$\mathbf{a}^* = \{\mathrm{diag}(\mathbf{p}_t)\}^{-1}\boldsymbol{\Omega}_t^{-1}\tilde{\boldsymbol{\mu}}_t,$$

which can be estimated by

$$\hat{\mathbf{a}}_n^* = \{\mathrm{diag}(\mathbf{p}_t)\}^{-1}\hat{\boldsymbol{\Omega}}_n^{-1}\{\hat{\boldsymbol{\mu}}_n - (1+r_t)\mathbf{e}\}.$$

Proposition

Under **A0-A1***, we have, conditional on* $\mathbf{p}_t$*,*

$$\hat{\mathbf{a}}_n^* \to \mathbf{a}^*, \quad a.s.,$$

$$\sqrt{n}\{\hat{\mathbf{a}}_n^* - \mathbf{a}^*\} \xrightarrow{d} \mathcal{N}\left(0, \{diag(\mathbf{p}_t)\}^{-1}\boldsymbol{\Omega}^{-1}\{diag(\mathbf{p}_t)\}^{-1}(1+S_t^2) + \mathbf{a}^*\mathbf{a}^{*\prime}\right).$$

Proof (I)

We have

$$\boldsymbol{\omega}_{\mathbf{a}}^{*} = \boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}, \quad \hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} = \mathrm{diag}(\mathbf{p}_t)\hat{\mathbf{a}}_n^{*} = \hat{\boldsymbol{\Omega}}_n^{-1}\{\hat{\boldsymbol{\mu}}_n - (1+r_t)\mathbf{e}\}.$$

It suffices to show that

$$\sqrt{n}\{\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*}\} \xrightarrow{d} \mathcal{N}\left(0, \boldsymbol{\Omega}^{-1}(1+S_t^2) + \boldsymbol{\omega}_{\mathbf{a}}^{*}\boldsymbol{\omega}_{\mathbf{a}}^{*\prime}\right).$$

Let $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - (1+r_t)\mathbf{e}$ et $\tilde{\boldsymbol{\mu}}_n = \hat{\boldsymbol{\mu}}_n - (1+r_t)\mathbf{e}$. We have

$$\begin{aligned}
\sqrt{n}\{\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*}\} &= \sqrt{n}(\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n - \boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}) \\
&= \boldsymbol{\Omega}^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n - \tilde{\boldsymbol{\mu}}) + \sqrt{n}(\hat{\boldsymbol{\Omega}}_n^{-1} - \boldsymbol{\Omega}^{-1})\tilde{\boldsymbol{\mu}}_n \\
&\stackrel{d}{=} \boldsymbol{\Omega}^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n - \tilde{\boldsymbol{\mu}}) + \sqrt{n}(\hat{\boldsymbol{\Omega}}_n^{-1} - \boldsymbol{\Omega}^{-1})\tilde{\boldsymbol{\mu}}.
\end{aligned}$$

Proof (II)

The two terms in the r.h.s. are asymptotically independent. We have

$$\mathbf{\Omega}^{-1}\sqrt{n}(\tilde{\boldsymbol{\mu}}_n - \tilde{\boldsymbol{\mu}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbf{\Omega}^{-1}\right).$$

We also have

$$\begin{aligned} \sqrt{n}(\hat{\mathbf{\Omega}}_n^{-1} - \mathbf{\Omega}^{-1})\tilde{\boldsymbol{\mu}} &= \hat{\mathbf{\Omega}}_n^{-1}\sqrt{n}(\mathbf{\Omega} - \hat{\mathbf{\Omega}}_n)\mathbf{\Omega}^{-1}\tilde{\boldsymbol{\mu}} \\ &\overset{d}{=} \mathbf{\Omega}^{-1}\sqrt{n}(\mathbf{\Omega} - \hat{\mathbf{\Omega}}_n)\mathbf{\Omega}^{-1}\tilde{\boldsymbol{\mu}}. \end{aligned}$$

Moreover,

$$\sqrt{n}(\mathbf{\Omega} - \hat{\mathbf{\Omega}}_n)\mathbf{\Omega}^{-1}\tilde{\boldsymbol{\mu}} \xrightarrow{d} \mathcal{N}\left(0, (\tilde{\boldsymbol{\mu}}'\mathbf{\Omega}^{-1}\tilde{\boldsymbol{\mu}})\mathbf{\Omega} + \tilde{\boldsymbol{\mu}}\tilde{\boldsymbol{\mu}}'\right).$$

□

Test of equality of performance of two portfolios

Recall that, the performance of a portfolio is measured via the Sharpe ratio

$$S_t(\mathbf{a}) = \frac{\boldsymbol{\omega}_{\mathbf{a}}' \tilde{\boldsymbol{\mu}}_t}{(\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Omega}_t \boldsymbol{\omega}_{\mathbf{a}})^{1/2}}$$

where $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}$ and

$$\boldsymbol{\omega}_{\mathbf{a}} = \mathrm{diag}(\mathbf{p}_t)\mathbf{a} = \begin{pmatrix} a_1 p_{1t} \\ \vdots \\ a_N p_{Nt} \end{pmatrix}.$$

Test of equality of performance of two portfolios

Let two portfolios $\mathbf{a}$ and $\mathbf{b}$.

We wish to test, at time t,

$$H_0: \quad S_t(\mathbf{a}) = S_t(\mathbf{b}).$$

Proposition

Under **A0-A1**, *we have, conditional on* $\mathbf{p}_t$,

$$\sqrt{n}\{\hat{S}_{t,n}(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b}) - S_t(\mathbf{a}) + S_t(\mathbf{b})\}$$
$$\xrightarrow{d} \quad \mathcal{N}\left(0, \frac{1}{2}\{S_t(\mathbf{a}) - S_t(\mathbf{b})\}^2 + 2(1-\rho_{\mathbf{ab}}) + S_t(\mathbf{a})S_t(\mathbf{b})(1-\rho_{\mathbf{ab}}^2)\right),$$

where $\rho_{\mathbf{ab}} = \dfrac{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}{\sqrt{(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}_t\boldsymbol{\omega}_{\mathbf{a}})(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})}} = Corr_t(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{y}_{t+1}, \boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{y}_{t+1}).$

Proof (I)

Recall that

$$\hat{S}_{t,n}(\mathbf{a}) = \frac{\boldsymbol{\omega}'_{\mathbf{a}}\tilde{\boldsymbol{\mu}}_n}{\{\boldsymbol{\omega}'_{\mathbf{a}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}}\}^{1/2}}, \qquad \tilde{\boldsymbol{\mu}}_n = \hat{\boldsymbol{\mu}}_n - (1+r_t)\mathbf{e}.$$

We can write $\hat{S}_{t,n}(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b})$ as a differentiable function f of the estimators $\hat{\boldsymbol{\mu}}_n, \boldsymbol{\omega}'_{\mathbf{a}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}}$ and $\boldsymbol{\omega}'_{\mathbf{b}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{b}}$.

$$\hat{S}_{t,n}(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b}) = f\left(\tilde{\boldsymbol{\mu}}_n,\ \boldsymbol{\omega}'_{\mathbf{a}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}},\ \boldsymbol{\omega}'_{\mathbf{b}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{b}}\right)$$

with

$$f(\boldsymbol{\mu}, z_1, z_2) = \left(\frac{\boldsymbol{\omega}'_{\mathbf{a}}}{\sqrt{z_1}} - \frac{\boldsymbol{\omega}'_{\mathbf{b}}}{\sqrt{z_2}}\right)\boldsymbol{\mu}.$$

Similarly

$$S_t(\mathbf{a}) - S_t(\mathbf{b}) = f\left(\tilde{\boldsymbol{\mu}},\ \boldsymbol{\omega}'_{\mathbf{a}}\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}},\ \boldsymbol{\omega}'_{\mathbf{b}}\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}\right).$$

Proof (II)

We have

$$\sqrt{n}\begin{pmatrix} \boldsymbol{\omega}_{\mathbf{a}}'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{a}} \\ \boldsymbol{\omega}_{\mathbf{b}}'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{b}} \end{pmatrix} \xrightarrow{d} \mathbf{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2 & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \\ 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 & 2(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \end{pmatrix}\right).$$

Indeed,

$$\begin{aligned}
& \mathrm{Cov}_{as}\left(\boldsymbol{\omega}_{\mathbf{a}}'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{a}}, \boldsymbol{\omega}_{\mathbf{b}}'(\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{b}}\right) \\
= \quad & \boldsymbol{\omega}_{\mathbf{a}}'\mathrm{Cov}_{as}\left((\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\boldsymbol{\omega}_{\mathbf{a}}, (\hat{\boldsymbol{\Omega}}_n - \boldsymbol{\Omega})\right)\boldsymbol{\omega}_{\mathbf{b}})\boldsymbol{\omega}_{\mathbf{b}} \\
= \quad & \boldsymbol{\omega}_{\mathbf{a}}'\left((\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})\boldsymbol{\Omega} + \boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\right)\boldsymbol{\omega}_{\mathbf{b}} \\
= \quad & (\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}) + \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}} \\
= \quad & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2.
\end{aligned}$$

Proof (III)

Using the asymptotic independence between $\tilde{\boldsymbol{\mu}}_n$ et $\hat{\boldsymbol{\Omega}}_n$, and the delta method we get the asymptotic law of

$$\begin{aligned} & \sqrt{n}(\hat{S}_{t,n}(\mathbf{a}) - S_t(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b}) + S_t(\mathbf{b})) \\ = \; & \sqrt{n}\{f(\tilde{\boldsymbol{\mu}}_n,\ \boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}},\ \boldsymbol{\omega}_{\mathbf{b}}'\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{b}}) - f(\tilde{\boldsymbol{\mu}},\ \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}},\ \boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})\} \\ \xrightarrow{d} \; & \mathbf{N}\left(0, \boldsymbol{A}\begin{pmatrix} \boldsymbol{\Omega} & 0 & 0 \\ 0 & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2 & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \\ 0 & 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 & 2(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \end{pmatrix}\boldsymbol{A}'\right), \end{aligned}$$

where

$$\boldsymbol{A} = \left[\frac{\partial f}{\partial \tilde{\boldsymbol{\mu}}'}, \quad \frac{\partial f}{\partial z_1}, \quad \frac{\partial f}{\partial z_2}\right]_{\substack{z_1 = \boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}} \\ z_2 = \boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}}$$

Proof (IV)

$$
\begin{aligned}
\boldsymbol{A} &= \left[\frac{\partial f}{\partial \tilde{\boldsymbol{\mu}}'}, \quad \frac{\partial f}{\partial z_1}, \quad \frac{\partial f}{\partial z_2}\right]_{\substack{z_1=\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}} \\ z_2=\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}} \\
&= \left[\frac{\boldsymbol{\omega}_{\mathbf{a}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}} - \frac{\boldsymbol{\omega}_{\mathbf{b}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}}, \quad \frac{-\boldsymbol{\omega}_{\mathbf{a}}'\tilde{\boldsymbol{\mu}}}{2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^{3/2}}, \quad \frac{\boldsymbol{\omega}_{\mathbf{b}}'\tilde{\boldsymbol{\mu}}}{2(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^{3/2}}\right] \\
&= \left[\frac{\boldsymbol{\omega}_{\mathbf{a}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}} - \frac{\boldsymbol{\omega}_{\mathbf{b}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}}, \quad \frac{-S_t(\mathbf{a})}{2\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}, \quad \frac{S_t(\mathbf{b})}{2\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}\right].
\end{aligned}
$$

Proof (V)

Thus the asymptotic variance :

$$
\begin{aligned}
& \left(\frac{\boldsymbol{\omega}_{\mathbf{a}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}} - \frac{\boldsymbol{\omega}_{\mathbf{b}}'}{\sqrt{\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}}\right) \boldsymbol{\Omega} \left(\frac{\boldsymbol{\omega}_{\mathbf{a}}}{\sqrt{\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}}} - \frac{\boldsymbol{\omega}_{\mathbf{b}}}{\sqrt{\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}}}\right) \\
+ \quad & \frac{S_t^2(\mathbf{a})}{4(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2} 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}})^2 + \frac{S_t^2(\mathbf{b})}{4(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2} 2(\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \\
- \quad & 2\frac{S_t(\mathbf{a})}{2\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{a}}} \frac{S_t(\mathbf{b})}{2\boldsymbol{\omega}_{\mathbf{b}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}}} 2(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Omega}\boldsymbol{\omega}_{\mathbf{b}})^2 \\
= \quad & \frac{1}{2}\{S_t(\mathbf{a}) - S_t(\mathbf{b})\}^2 + 2(1-\rho_{\mathbf{ab}}) + S_t(\mathbf{a})S_t(\mathbf{b})(1-\rho_{\mathbf{ab}}^2).
\end{aligned}
$$

Test

Note that by the Cauchy-Schwarz inequality, $\rho_{\mathbf{ab}} \in [-1,1]$, with equality to 1 iff $\boldsymbol{\omega}_{\mathbf{a}} = \boldsymbol{\omega}_{\mathbf{b}}$ (in this case the normal law is degenerate).

Corollary

Under the same assumptions, a test statistic for testing the equality of performance of portfolios $\mathbf{a}$ *and* $\mathbf{b}$ *at time* t *is*

$$\xi_n = \frac{n\{\hat{S}_{t,n}(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b})\}^2}{2(1-\hat{\rho}_{\mathbf{ab}}) + \hat{S}^2_{t,n}(\mathbf{a})(1-\hat{\rho}^2_{\mathbf{ab}})}, \text{ where } \hat{\rho}_{\mathbf{ab}} = \frac{\boldsymbol{\omega}'_{\mathbf{a}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{b}}}{\sqrt{(\boldsymbol{\omega}'_{\mathbf{a}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{a}})(\boldsymbol{\omega}'_{\mathbf{b}}\hat{\boldsymbol{\Omega}}_n\boldsymbol{\omega}_{\mathbf{b}})}}.$$

Under H_0 *and if* $\rho_{\mathbf{ab}} \neq 1$,

$$\xi_n \xrightarrow{d} \chi^2(1).$$

Proof

Under H_0,

$$\sqrt{n}\{\hat{S}_{t,n}(\mathbf{a}) - \hat{S}_{t,n}(\mathbf{b})\} \xrightarrow{d} \mathcal{N}\left(0, 2(1-\rho_{\mathbf{ab}}) + S_t^2(\mathbf{a})(1-\rho_{\mathbf{ab}}^2)\right)$$

and the conclusion follows, noting that $\hat{\rho}_{\mathbf{ab}}$ converges a.s. to $\rho_{\mathbf{ab}}$.

$\square$

In practice, we reject H_0 at the asymptotic level $\alpha \in]0,1[$ if

$$\xi_n > \chi^2_{1-\alpha}(1),$$

where $\chi^2_{1-\alpha}(1)$ is the $(1-\alpha)-$quantile of the $\chi^2(1)$ distribution.

Remarks

The assumption that $\rho_{\mathbf{ab}} \neq 1$ ensures that the statistic is well-defined, at least for n large enough.

Note that the Cauchy-Schwarz inequality reduces to an equality only when the vectors are colinear : thus $\rho_{\mathbf{ab}} = 1$ means that $\mathbf{a} = K\mathbf{b}$ for some constant K.

In particular, the test does not apply to efficient portfolios, because such portfolios have the same composition in risky assets up to the factor λ.

Testing the efficiency of a portfolio at time t

Efficient allocations (in value) are proportional to $\boldsymbol{\omega}_{\mathbf{a}}^* = \boldsymbol{\Omega}^{-1}\tilde{\boldsymbol{\mu}}$.

The assumption that a portfolio of composition $\boldsymbol{\omega}_{\mathbf{a}}$ at time t is efficient can thus be written

$$H_0: \quad \exists k \in \mathbb{R}, \quad \boldsymbol{\omega}_{\mathbf{a}}^* = k\boldsymbol{\omega}_{\mathbf{a}}.$$

The alternative assumption is

$$H_1: \quad \nexists k \in \mathbb{R}, \quad \boldsymbol{\omega}_{\mathbf{a}}^* = k\boldsymbol{\omega}_{\mathbf{a}}.$$

Recall that

$$\sqrt{n}\{\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \boldsymbol{\omega}_{\mathbf{a}}^*\} \xrightarrow{d} \mathcal{N}\left(0,\ \boldsymbol{\Sigma} := \boldsymbol{\Omega}^{-1}(1+S_t^2) + \boldsymbol{\omega}_{\mathbf{a}}^*\boldsymbol{\omega}_{\mathbf{a}}^{*\prime}\right).$$

Let $\hat{\boldsymbol{\Sigma}}_n = \hat{\boldsymbol{\Omega}}_n^{-1}(1+\hat{S}_{t,n}^2) + \hat{\boldsymbol{\omega}}_{\mathbf{a}}^*\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*\prime}$, with $\hat{S}_{t,n} = \sqrt{\tilde{\boldsymbol{\mu}}_n'\hat{\boldsymbol{\Omega}}_n^{-1}\tilde{\boldsymbol{\mu}}_n}$.

$\boldsymbol{\Sigma}$ is positive definite, thus $\hat{\boldsymbol{\Sigma}}_n$ is non-singular for sufficiently large n.

Test statistic

$$\begin{aligned}\xi_n^* &= \min_k n(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - k\boldsymbol{\omega}_{\mathbf{a}})'\hat{\boldsymbol{\Sigma}}_n^{-1}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - k\boldsymbol{\omega}_{\mathbf{a}}) \\ &= n(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k}\boldsymbol{\omega}_{\mathbf{a}})'\hat{\boldsymbol{\Sigma}}_n^{-1}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k}\boldsymbol{\omega}_{\mathbf{a}})\end{aligned}$$

where

$$\hat{k} = (\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Sigma}}_n^{-1}\boldsymbol{\omega}_{\mathbf{a}})^{-1}\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Sigma}}_n^{-1}\hat{\boldsymbol{\omega}}_{\mathbf{a}}^*$$

is the QGLS estimator of k in the regression of $\hat{\boldsymbol{\omega}}_{\mathbf{a}}^*$ on $\boldsymbol{\omega}_{\mathbf{a}}$. QGLS

Proposition

Under the previous assumptions,

$$\xi_n^* \xrightarrow{d} \chi^2(N-1) \qquad \textit{under } H_0$$

$$\xi_n^* \xrightarrow{a.s.} +\infty \qquad \textit{under } H_1.$$

Proof (I)

Under H_0, we have

$$\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \hat{k}\boldsymbol{\omega}_{\mathbf{a}}) = \sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*}) + \boldsymbol{\omega}_{\mathbf{a}}\sqrt{n}(k - \hat{k}).$$

We have (because $\boldsymbol{\omega}_{\mathbf{a}}^{*} = k\boldsymbol{\omega}_{\mathbf{a}}$)

$$k - \hat{k} = (\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Sigma}}_n^{-1}\boldsymbol{\omega}_{\mathbf{a}})^{-1}\boldsymbol{\omega}_{\mathbf{a}}'\hat{\boldsymbol{\Sigma}}_n^{-1}(\boldsymbol{\omega}_{\mathbf{a}}^{*} - \hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*}) := M_n'(\boldsymbol{\omega}_{\mathbf{a}}^{*} - \hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*}).$$

Thus

$$\begin{aligned}
\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \hat{k}\boldsymbol{\omega}_{\mathbf{a}}) &= (I_N - \boldsymbol{\omega}_{\mathbf{a}}M_n')\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*}) \\
&\overset{d}{=} \{I_N - \boldsymbol{\omega}_{\mathbf{a}}(\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\omega}_{\mathbf{a}})^{-1}\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Sigma}^{-1}\}\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*}) \\
&:= \{I_N - \boldsymbol{Q}\boldsymbol{\Sigma}^{-1}\}\underbrace{\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^{*} - \boldsymbol{\omega}_{\mathbf{a}}^{*})}_{\xrightarrow{d}\mathcal{N}(0,\ \boldsymbol{\Sigma})} \\
&\xrightarrow{d} \mathcal{N}\left(0,\ \boldsymbol{\Sigma} - 2\boldsymbol{Q} + \boldsymbol{Q}\boldsymbol{\Sigma}^{-1}\boldsymbol{Q}\right).
\end{aligned}$$

Proof (II)

Let

$$\mathbf{P} = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{Q} \boldsymbol{\Sigma}^{-1/2} = \frac{\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\omega}_{\mathbf{a}} \boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Sigma}^{-1/2}}{\boldsymbol{\omega}_{\mathbf{a}}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\omega}_{\mathbf{a}}}.$$

Note that $\mathbf{P}$ is an orthogonal projection matrix :

$$\mathbf{P} = \mathbf{P}' \text{ (symmetry)}, \quad \mathbf{P}^2 = \mathbf{P} \text{ (idempotence)}$$

Orthogonal projection on the **vector** $\boldsymbol{X} = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\omega}_{\mathbf{a}}$ (since $\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$).

We find that

$$\boldsymbol{\Sigma}^{-1/2} \sqrt{n} (\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k} \boldsymbol{\omega}_{\mathbf{a}}) \quad \xrightarrow{d} \quad \mathcal{N}\left(0, I_N - 2\mathbf{P} + \mathbf{P}^2\right) = \mathcal{N}\left(0, I_N - \mathbf{P}\right).$$

It follows that

$$\hat{\boldsymbol{\Sigma}}_n^{-1/2} \sqrt{n} (\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k} \boldsymbol{\omega}_{\mathbf{a}}) \quad \xrightarrow{d} \quad \mathcal{N}\left(0, I_N - \mathbf{P}\right)$$

The variance-covariance matrix is the orthogonal projection matrix on **the orthogonal space** of the vector $\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\omega}_{\mathbf{a}}$: it has rank $N-1$.

Proof (III)

If $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q})$ where $\boldsymbol{Q}$ is a symmetric projection matrix of rank K, then $\boldsymbol{X}'\boldsymbol{X} \sim \chi^2(K)$.

Why ?

There exists a diagonal matrix $\boldsymbol{D}$ and an orthogonal matrix $\boldsymbol{S}$ (i.e. $\boldsymbol{S}\boldsymbol{S}' = \boldsymbol{S}'\boldsymbol{S} = \boldsymbol{I}$) such that

$$\boldsymbol{D} = \boldsymbol{S}\boldsymbol{Q}\boldsymbol{S}'.$$

Thus

$$\boldsymbol{D}^2 = \boldsymbol{S}\boldsymbol{Q}\boldsymbol{S}'\boldsymbol{S}\boldsymbol{Q}\boldsymbol{S}' = \boldsymbol{D}.$$

There are K terms equal to 1 on the diagonal of $\boldsymbol{D}$, the remaining terms being 0.
Let $\boldsymbol{Y} = \boldsymbol{S}\boldsymbol{X}$. Then $\boldsymbol{Y} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{S}\boldsymbol{Q}\boldsymbol{S}') = \mathcal{N}(\mathbf{0}, \boldsymbol{D})$. We have

$$\boldsymbol{X}'\boldsymbol{X} = \boldsymbol{Y}'\boldsymbol{Y} = \sum_{i=1}^{K} Y_i^2 \sim \chi^2(K).$$

Proof (IV)

We have seen that

$$\boldsymbol{Z}_n := \hat{\boldsymbol{\Sigma}}_n^{-1/2}\sqrt{n}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k}\boldsymbol{\omega}_{\mathbf{a}}) \quad \xrightarrow{d} \quad \mathcal{N}(0, I_N - \mathbf{P}), \quad \mathrm{rank}(I_N - \mathbf{P}) = N - 1.$$

Thus

$$\xi_n^* \quad = \quad \boldsymbol{Z}_n'\boldsymbol{Z}_n \sim \chi^2(N-1).$$

Hence the conclusion under H_0.

Under H_1 we have, for $k = (\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\omega}_{\mathbf{a}})^{-1}\boldsymbol{\omega}_{\mathbf{a}}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\omega}_{\mathbf{a}}^*$,

$$\frac{1}{n}\xi_n^* = (\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k}\boldsymbol{\omega}_{\mathbf{a}})'\hat{\boldsymbol{\Sigma}}_n^{-1}(\hat{\boldsymbol{\omega}}_{\mathbf{a}}^* - \hat{k}\boldsymbol{\omega}_{\mathbf{a}}) \to (\boldsymbol{\omega}_{\mathbf{a}}^* - k\boldsymbol{\omega}_{\mathbf{a}})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\omega}_{\mathbf{a}}^* - k\boldsymbol{\omega}_{\mathbf{a}}).$$

This term is non zero under H_1, since $\boldsymbol{\Sigma}^{-1}$ is positive definite.

$\square$

Rejection region

In practice, H_0 is rejected at the asymptotic level $\alpha \in]0,1[$ if

$$\xi_n^* > \chi^2_{1-\alpha}(N-1).$$

Remark : It can be shown that

$$\xi_n^* = n\frac{\hat{S}_t^2 - \hat{S}_{t,n}^2(\mathbf{a})}{1+\hat{S}_t^2}.$$

Reminders on regression

In the regression model

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{U}, \quad E(\boldsymbol{U} \mid \boldsymbol{X}) = \boldsymbol{0}, \quad V(\boldsymbol{U} \mid \boldsymbol{X}) = \boldsymbol{\Omega} \text{ (positive definite)}$$

where $\boldsymbol{Y} \in \mathbb{R}^N$, $\boldsymbol{X}$ is a $N \times K$ matrix, the OLS estimator of $\boldsymbol{\beta} \in \mathbb{R}^K$ solves

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{Y} - \boldsymbol{X\beta}\|^2 = \arg\min_{\boldsymbol{\beta}} (\boldsymbol{Y} - \boldsymbol{X\beta})'(\boldsymbol{Y} - \boldsymbol{X\beta})$$

and the solution is (if $\boldsymbol{X}'\boldsymbol{X}$ has full rank)

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}.$$

$\boldsymbol{X}\widehat{\boldsymbol{\beta}}$ is the **orthogonal** projection of $\boldsymbol{Y}$ on the columns of $\boldsymbol{X}$, with projection matrix

$$\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'.$$

Reminders on regression

A more efficient estimator is the Generalized Least-Squares (GLS) estimator

$$\widehat{\boldsymbol{\beta}}_{GLS} = (\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{Y}$$

which is the solution of the minimization problem :

$$\widehat{\boldsymbol{\beta}}_{GLS} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{Y}-\boldsymbol{X}\boldsymbol{\beta}\|_{\Omega}^{2} = \arg\min_{\boldsymbol{\beta}} (\boldsymbol{Y}-\boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{\Omega}^{-1}(\boldsymbol{Y}-\boldsymbol{X}\boldsymbol{\beta}).$$

$\boldsymbol{X}\widehat{\boldsymbol{\beta}}_{GLS}$ is an **oblique** projection of $\boldsymbol{Y}$ on the columns of $\boldsymbol{X}$, with projection matrix

$$\boldsymbol{P}^{*} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}^{-1}.$$

A Quasi-GLS (QGLS) estimator of $\boldsymbol{\beta}$ is

$$\widehat{\boldsymbol{\beta}}_{QGLS} = (\boldsymbol{X}'\widehat{\boldsymbol{\Omega}}_{n}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\widehat{\boldsymbol{\Omega}}_{n}^{-1}\boldsymbol{Y}$$

where $\widehat{\boldsymbol{\Omega}}_{n}$ is a consistent estimator of $\boldsymbol{\Omega}$.

Return